

Climbing the Ladder: Automating Data Pipelines and Reproducing Baselines in Low-Resource Settings

Donato Meoli
Dipartimento di Informatica, Università di Pisa
Largo B. Pontecorvo 3, 56127 Pisa – Italy
`donato.meoli@outlook.com`

Eryk Wdowiak
Arba Sicula
`eryk@wdowiak.me`

September 28, 2026

Abstract

The massively multilingual translation systems that now include Sicilian learned it from noisy, non-standardised web text. To obtain better translation quality, we redevelop the small neural machine translation (NMT) system that Wdowiak [35] customized for Sicilian. We rebuild on maintained software the extraction of the parallel text from the PDFs of the bilingual journal *Arba Sicula*, whose sentences are aligned by a language-agnostic sentence encoder with thresholds tuned against an existing hand alignment and then filtered for the typical errors of automatic alignment. Furthermore, the tokenization and the desinence-biased subword segmentation of the original system are ported to a Perl-free implementation, and all the results are measured on a frozen test set drawn from the journal. We compare small Transformers trained from scratch with the pretrained NLLB model adapted by low-rank fine-tuning, in both directions and with Italian as a closely related bridge language, and we adapt to the pretrained model the reverse-training recipe of the original system, i.e., a stage on forward-scored back-translations of the web Sicilian followed by a stage on the curated literary text. The pretrained model is by far the better of the two, and the staged training improves the translation into Sicilian more than the translation from it. Yet, both directions remain well below the results that the original recipe obtains on its test set, and a multilingual adapter degrades the directions on which it was not trained, which limits its use as a general multilingual model.

1 Introduction

Languages without an army and navy often lack a standard orthography. Without specific spelling rules and conventions, written texts in these languages often exhibit great variability (sometimes even within a local dialect), which inhibits the ability of conventional metrics to assess model performance [1]. But even languages like Sicilian – where published dictionaries have presented a consistent orthography for several hundred years [14, 15, 18, 31] – can exhibit extreme orthographic diversity which (when incorporated into a training dataset) prevents a model from learning how to generate consistent text.

During the 13th century, the Sicilian School of Poets at the court of Emperor Frederick II created the first literary standard in Italy. Their poetry inspired Dante to write in his native Florentine, which later became today’s Italian language. Following in their footsteps, poets

like Antonio Veneziano (1543-1593), Giovanni Meli (1740-1815), Domenico Tempio (1750-1821), Ignazio Buttitta (1899-1997) and Salvatore Camilleri (1921-2021) continued the Sicilian literary tradition, writing in a fairly consistent manner. But writing is different from speaking, so Giuseppe Pitrè (1841-1916) developed a popular tradition which recorded the common people in their own words and in the sound of their own words.

One can hear echoes of the popular tradition in the daily speech of millions of people in Sicily, Calabria and Puglia, as well as in diaspora communities abroad. And one can find echoes of the literary tradition in books, journals and pages of the internet. Nonetheless, the publicly available Sicilian language text often exhibits extreme orthographic diversity which once prevented practitioners from developing datasets of Sicilian-language parallel text and once caused Sicilian to be absent from every major translation service.

Sicilian is now covered by massively multilingual models such as NLLB-200 [16]; however, its training text was mined from non-standard web sources, and the model has in effect learned a non-standard orthography (cf. section 4). Hence, the problem for Sicilian has moved from the absence of a model to the quality of the data on which it is trained.

High-quality material exists, especially in the bilingual literary journal *Arba Sicula* (published since 1979), which Wdowiak [35] used to create the first Sicilian neural machine translation (NMT) system. His work demonstrated that it’s possible to develop a fairly good translation model in low-resource settings, but it was a labor-intensive task. He aligned sentences by hand using a recipe tailored to the low-resource, morphologically rich setting.

This work rebuilds that effort with maintained, mostly CPU-only tooling and seeks to obtain better quality with modern pretrained models.

That system [34, 35] followed the low-resource recipe of Sennrich and Zhang [25], i.e., a small and heavily regularized Transformer [32] with a small subword vocabulary [27], and biased the segmentation toward the desinences of Sicilian grammar; its later reverse-training strategy [36] builds a pretrained model of Sicilian in stages from synthetic and web text before the fine-tuning on the literary one. Parallel text is usually aligned automatically with dictionary-based tools such as hunalign or, more recently, with multilingual sentence encoders such as LaBSE [6] and LASER3 [7], which were used to mine WikiMatrix [23] and the Sicilian data of NLLB. Vecalign [29] combines such embeddings with a dynamic program, and collections such as OPUS [30] gather the resulting corpora. Pretrained multilingual models can be adapted to a specific pair at a small cost by low-rank adapters [9], and allow one to exploit closely related high-resource languages as a bridge [2, 5, 10, 37]; also, back-translation of monolingual text [26] is the standard way of adding in-domain synthetic data. The reader is referred to Koehn and Knowles [11] for a discussion of why low-resource translation remains difficult.

In particular, we extract the parallel text of the journal automatically, with a LaBSE-based aligner whose thresholds are tuned against the hand alignment of the second author and whose output is filtered for the errors that his review of it brought to light; the aligned text and his hand-aligned sheets are then used together, since the latter cover only a fraction of the pairs of each issue. Furthermore, we port to Python the tokenization and the desinence-biased segmentation of the original system, and add a lemma source factor [24] to the from-scratch baseline. Finally, we fine-tune NLLB-200 with LoRA in both directions and with Italian as a bridge, and we replace the custom pretraining of the reverse-training recipe with the pretraining of NLLB, keeping its second stage (forward-scored back-translations of the web Sicilian) and its last one (the curated literary text). We remark that the published results of the original system were measured on a different test set, and that the comparison with them is therefore indirect (cf. section 8).

The structure of the paper is as follows. Section 2 recalls the models on which all our systems are based. Section 3 describes the extraction, the alignment and the cleaning of the parallel text and the assembly of the datasets, while section 4 discusses the orthography of the Sicilian data and the reason why a multilingual encoder can align it at all. Tokenization

and subword segmentation are the subject of section 5, the models and the staged training of section 6, and the evaluation protocol of section 7; the results are reported in section 8, and section 9 concludes.

2 Background

We now recall the model on which all our systems are based. NMT treats translation as a conditional language model: an encoder reads the source sentence $x = (x_1, \dots, x_n)$, and a decoder factorizes the probability of the target sentence y from left to right as

$$p_{\theta}(y \mid x) = \prod_t p_{\theta}(y_t \mid y_{<t}, x) ,$$

where the parameters θ are trained by maximizing the log-likelihood of a parallel corpus, and the translation is obtained by beam search. Its standard parameterization is the Transformer [32], which replaces recurrence with the scaled dot-product attention $\text{Attention}(Q, K, V) = \text{softmax}(QK^{\top}/\sqrt{d_k})V$ (where Q , K and V are the query, key and value matrices, and d_k is the dimension of the keys), computed in several parallel heads.

With little data the default Transformer overfits, and hence the low-resource recipe of Senrich and Zhang [25] prescribes a smaller model with high dropout [28] and a small subword vocabulary. Byte-Pair Encoding (BPE) [27] gives an open vocabulary and shares statistics across the forms of a word, which is especially useful for a morphologically rich language such as Sicilian. Further improvements usually come from deeper models, when the data allow them [13], and from linguistic input features [24], which we use in section 6.

3 Data Pipeline

3.1 Extraction from PDFs

We read the text layer of the *Arba Sicula* PDFs with PyMuPDF, which avoids the pdflatex round trip of the original pipeline. However, the quality of this layer depends on the issue. In fact, the PDFs of the issues up to AS18, and of AS21, are scans of the paper copies whose OCR text carries no accent over the vowels; from AS19 (1998) onward they were produced from the original computer files. This date also marks (most likely) an improvement of the Sicilian itself, since 1998 is the year of Privitera’s *Beginner’s Sicilian* [22], the first Sicilian grammar published after the one of Pitre [19], which was followed by the more comprehensive grammar of Bonner [3]; hence, the scanned issues are kept out of the training data (they are used only for the complete extraction of the archive mentioned below). Some of the born-digital issues (e.g., AS22 and AS23) contain spaces of zero width inside the words, which we detect from the glyph positions and remove before the text is read.

Each page is assigned a language according to its share of Sicilian and English stop-words, and pages with too few tokens are left out. Since the journal usually sets Sicilian and English on facing pages, each Sicilian page is compared with the English page on either side, and the pairs are chosen greedily by decreasing LaBSE similarity (cf. section 3.2), each page being used at most once; this corrects the pairings that a fixed left/right rule gets wrong wherever a page is mislabelled or the order of the two languages is swapped. Pages are then segmented into sentences with `pysbd` (with the Italian rules for Sicilian), after the running headers and page numbers have been removed; fragments with fewer than two words or mostly in capitals are dropped, and the words hyphenated at the end of a line are rejoined.

3.2 Sentence alignment

We embed each sentence with LaBSE [6], a language-agnostic dual encoder, and build the cosine-similarity matrix

$$S_{ij} = \frac{\langle u_i, v_j \rangle}{\|u_i\| \|v_j\|},$$

where u_i and v_j are the embeddings of the i -th Sicilian and of the j -th English sentence. For a pair of pages, the similarity is the larger of the mean row maximum and the mean column maximum of S , and the pair is kept if it is at least τ_{page} (with at least two sentences on each page). Consecutive spreads are then chained into blocks, so that a sentence cut by a page break is mended when it has no closing punctuation and the next page starts in lowercase. Within a block, the sentences are aligned by a monotonic dynamic program over S with moves 1–1, 1–2 and 2–1, whose gain is the mean cosine of the sentences involved, and moves 1–0 and 0–1, which cost a penalty of 0.5; the program is banded to the facing page and its neighbors (and falls back to the unbanded one if the band admits no path). An aligned span is kept if its mean cosine is at least τ_{sent} . This is in the spirit of Vecalign [29], but requires neither a bilingual dictionary nor hunalign.

Both thresholds were chosen on a grid (0.50, 0.55, 0.58 and 0.62 for τ_{page} , 0.35, 0.40, 0.45 and 0.50 for τ_{sent}) against the hand alignment of the second author for the issues AS41 and AS42, which gave $(\tau_{\text{page}}, \tau_{\text{sent}}) = (0.50, 0.40)$. On that gold the aligner reproduces 678 pairs exactly (recall 0.53, precision 0.57); these figures are somewhat pessimistic, since a pair that is correct but segmented differently from the hand alignment counts as an error, and since the hand alignment favored, by its author’s account, the pairs that were easier to align. Each pair records the printed Sicilian and English page and its LaBSE similarity, which allows one to check it against the printed issue.

3.3 Cleaning

A review of the aligned text by the second author showed that the errors of the aligner are easy to recognize, and we filter them as follows. First, the residue of the page layout is removed from the sentences (the InDesign slug of AS43 and the “Return to the TOC” links). Then, a pair is dropped if (i) the longer side has more than 1.6 times the words of the shorter one (when it has at least five words), (ii) it contains a table-of-contents leader, (iii) it contains two or more characters typical of OCR noise, or (iv) more than 25% of its lowercase English words are not in an English dictionary. Note that the bound in (i) was first 2.0; the pairs between 1.6 and 2.0 have a mean LaBSE similarity of 0.59, against 0.73 for those below 1.6, which in our view justifies the tighter bound. After cleaning and deduplication, the whole digital archive of the journal gives 15,465 pairs.

3.4 Datasets

Two training sets are used (Table 1). Our first set, on which the from-scratch models and the first NLLB experiments were trained, joins our first extraction of 44 issues (14,245 pairs, before the cleaning of section 3.3) with the “Good-Sicilian-in-NLLB” set curated by the second author, i.e., sentences mined with LASER3 [7, 16] that were rescored and filtered for quality and for Corsican contamination. Its it–scn part is the WikiMatrix subset [23], with a small curated “Good-Sicilian-from-WikiMatrix” set.

Our second set, used by the staged training of section 6.3, replaces the web text with curated sources. These are (i) the hand-aligned sheets of the second author, i.e., Dieli’s translation of Pitre’s fables and 23 issues of the journal (scn–en), Cipolla’s translations of Meli (scn–it), and the trilingual exercises of the grammars of Bonner and Cipolla, (ii) our alignment of all the born-digital issues, since the sheets cover only 10 to 20% of the pairs of an issue and the two

Table 1: Datasets (number of sentence pairs). “Web” is the first training set of section 3.4, “curated” the second one; the test set is shared.

set	direction	pairs	sources
web	scn-en	27,386	Arba Sicula (13,538) + Good-Sicilian-in-NLLB (15,848)
	it-scn	9,389	WikiMatrix
curated	scn-en	30,980	hand-aligned sheets, Arba Sicula, Napizia
	scn-it	10,427	hand-aligned sheets, Napizia
	scn (mono)	14,360	Cipolla’s textbook
test	scn-en	1,000	Arba Sicula
	it-scn	1,000	WikiMatrix

sources are therefore complementary, and (iii) the parallel pages of the Napizia magazine, of the Napizia site and of the Young Sicilian Manifesto, from which one text under copyright was excluded. Furthermore, the sheets provide 14,360 monolingual Sicilian sentences (Cipolla’s 2021 textbook) and a validation set of 121 trilingual lines. Its Sicilian side is normalized as in section 5, and the exact repeats are removed.

A validation and a test set of 1,000 pairs each were drawn once at random from our first extraction of *Arba Sicula*, which provides the literary standard that FLORES does not, among the pairs with 3 to 80 words per side and a length ratio between 1/3 and 3, and then frozen; hence, all the results of section 8 are computed on the same test set. Of the two validation sets, the one of 1,000 pairs is used with the web set, and the 121 lines of the sheets with the curated one. Each training set is then guarded against it: a pair is removed if its Sicilian side is a test or validation sentence (in raw or normalized form), or contains a test sentence of six or more words. This guard removed 794 pairs from the second set (778 because of the test set and 16 because of the validation set), some of them from the hand-aligned sheets, which cover the same issues as the test set; without it, the results of the staged training would have been inflated.

4 Standardization

Although Sicilian does not have a single official orthography, a common literary standard has existed for several hundred years and provides a rough consistency, which we can refine because the majority of the Sicilian sentences in the NLLB dataset were written in that standard.

Nonetheless, approximately one-third of the Sicilian sentences that the NLLB team collected were written in local dialect. Some of the sentences were written in Corsican. And very few of the Sicilian sentences in the NLLB dataset were paired with a valid translation. We suspect that these data quality issues were caused by seed data which is inconsistent with the literary standard.

4.1 Why Sicilian lives in the Romance neighbourhood

A pretrained multilingual encoder needs no explicit Sicilian support to *read* Sicilian. The representations it learns are not a shared interlingua that maps every language into one abstract, universal meaning space; instead they cluster languages by linguistic similarity [12] and organise around surface-form cues [33]. For Sicilian and Italian, which are close on both counts, the two accounts agree: Sicilian lands in the Italian–Romance region either way.

The surface form of literary Sicilian is close to Italian, so reading it works well. Retrieval on the multi-parallel FLORES-200 devtest [16] confirms it: LaBSE retrieves the correct English translation of a Sicilian sentence in 99% of the cases (precision@1), one point below Italian

(100%), which it supports officially. Furthermore, if one discounts English (the pivot of LaBSE, to which all languages are close), the nearest translation of a Sicilian sentence is Italian in 18% of the cases, more than twice as often as French (8%), the next Romance language.

The seed-data failure, however, is in the *opposite* direction – retrieving the Sicilian translation of an English sentence. Romance languages share a lot of vocabulary and grammar, so when a model retrieves a Sicilian, Italian or Corsican translation of an English sentence, the translation that it selects will reflect the surface-form cues in its training data. Sometimes the NLLB seed data’s surface form reflects the literary language in Sicilian dictionaries, but other times it reflects local pronunciations, which (we suppose) is what caused their model to frequently retrieve Corsican-language sentences when assembling the NLLB’s Sicilian corpus.

Since multilingual models encode script and surface form more than deeper linguistic structure [33], the two directions behave differently for the same reason: an Italianate surface form retrieves the right English sentence, whereas seed data that look like Corsican lead the model to generate the wrong Sicilian. Hence, the proximity to Italian cuts both ways. It allows our LaBSE-based aligner (cf. section 3.2) to work on a language that LaBSE does not officially support; however, it gives no protection against contamination by neighboring languages when a corpus is assembled by transfer only.

How easily a text is aligned depends, in turn, on how standard it is. We computed, text by text, the LaBSE cosine of the hand-aligned gold pairs of AS41 and AS42: standard literary prose (Cipolla’s *Don Chisciotti*) has a mean cosine of 0.76, with 91% of its pairs above τ_{sent} , while dialect poetry has a mean of 0.64, with 80% of its pairs above it. Hence, the non-standard material is both the larger source of noise in the collected data and the harder to recover, and curating the data toward the literary standard should improve quality on both counts. For poetry, the second author suggested merging short consecutive verses into longer units before scoring. We simulated it on the poetry pairs of the gold: merging up to three verses raises the mean cosine by about 0.11 and the share above the threshold by 15 points (to 99%), although part of this gain is mechanical, since the merged units are fewer, longer and more distinctive. This corrective is not yet part of the production aligner.

For a translation model, we need sentences written in the Sicilian literary language and we need translations of those sentences. So we back-translated all of the Sicilian sentences into English and Italian, then we scored the resulting pair on the task of translation into Sicilian. Because a sentence written in standard Sicilian will score better than a sentence written in local dialect, forward-scoring the back-translations allows us to identify the sentences written in standard Sicilian. This method yielded 549,486 back-translations to simulate English-to-Sicilian translation and 689,840 to simulate Italian-to-Sicilian translation.

5 Tokenization and Subword Segmentation

Because the majority of the Sicilian sentences in the NLLB dataset were written in a common literary standard, we can refine them for consistency. We first normalize the text toward the standard established by Cipolla and augmented by Wdowiak for MT clarity, i.e., strict L on the articles (*lu/la/li*), h on *aviri* (*haiu, havi, hannu*), CI for *çi* (*çiuri*→*ciuri*) and R retained before a clitic (*farlu*). Our normalizer applies the documented spelling rules (e.g., *cchiù*→*chiù*, *io/iu/eu*→*jo*, *non*→*nun*) at a “std” level, and further uncontracts and ASCII-folds the text at a “full” level. Since the tokenizer of NLLB was trained on un-normalized web Sicilian, it is not obvious that normalization helps the pretrained model; hence, it was measured on *scn*→*en*, with the same English references for the three variants. BLEU is 30.60 on raw text, 30.98 with “std” and 30.98 with “full” (chrF 56.31, 56.63 and 56.50, respectively). Thus, the light “std” normalization gives a small but consistent improvement (and the best chrF), while the “full” one gives nothing more, which is consistent with a pretrained tokenizer that expects un-normalized input. We therefore adopt “std” for all the later experiments.

For the from-scratch models, the Napizia tokenizer was reimplemented in Python (its output coincides with that of the original Perl code): it ASCII-folds the accents and uncontracts the Sicilian forms (e.g., $\hat{o} \rightarrow a$ *lu*, $\hat{d}\hat{u} \rightarrow di$ *lu*), which reduces sparsity. Then, the joint scn-en BPE [27] is biased toward the textbook desinenes catalogued in the resources that Wdowiak assembled to standardize Sicilian (Diel’s *Sicilian Vocabulary*, Cipolla’s *Mparamu lu sicilianu* [4], Bonner’s grammar [3] and the *Chiù dâ Palora* dictionary), which are appended to the Sicilian side when the merges are learned. This keeps the morphological endings as units; for instance, Wdowiak’s sentence *Carinisi are dogs!* is segmented as *car@@ in@@ isi are dogs !*. In his experiments this reduced the vocabulary from 14,000 to 2,000 units and gave the largest improvement in BLEU. We call this combination of tokenizer and biased segmentation “lever B” in the following.

6 Models

6.1 From-scratch baseline

The from-scratch baseline is a small Transformer trained on CPU with Sockeye-3 [8]: 3 layers, model size 256, 4 attention heads, feed-forward size 1024, a vocabulary shared by the two languages and a joint BPE of 4000 merges, i.e., about 6.6M parameters. Dropout is 0.4 on the embeddings and 0.2 elsewhere, with label smoothing 0.1, and the model is trained with Adam (learning rate $2 \cdot 10^{-4}$, batches of 1024 words) for at most 30 epochs, with early stopping after 8 checkpoints without improvement. On top of it we add, in this order, lever B, more parallel text and a linguistic input feature (“lever D”). The latter is a lemma source factor [24] summed into the source embedding as

$$e_j = E_{\text{word}}(w_j) + E_{\text{lemma}}(\ell_j) ,$$

where w_j is the j -th source word and ℓ_j its lemma; the lemma vocabulary is small (about 900 entries), and therefore generalizes across the inflected forms of a word.

6.2 Pretrained model

Our pretrained model is NLLB-200 [16], a massively multilingual encoder–decoder that supports Sicilian (**scn_Latn**) and selects the target language through the first token forced on the decoder. We adapt its 1.3B variant with LoRA [9]: the pretrained weights W_0 of the query, key, value and output projections of the attention are frozen, and a low-rank update is learned, which gives $W = W_0 + \frac{\alpha}{r}BA$ with $r=32$, $\alpha=64$ and dropout 0.05 on the adapter (about 18.9M trainable parameters, i.e., 1.4% of the model). As usual, the base model is kept in fp16 and the adapters in fp32; training uses a learning rate of $2 \cdot 10^{-4}$ and sentences of at most 128 tokens, and decoding uses beam search with 5 beams. One adapter is trained on both directions at once (scn→en and en→scn), and one on four (scn↔en and it↔scn). For the back-translation experiment of section 8, we translate into Sicilian with the bidirectional adapter the 7,592 English sentences left unaligned by the first extraction, keep the 7,498 synthetic pairs whose length ratio is between 0.5 and 2 and whose Sicilian side is new, and train on them and on the web set a further adapter for scn→en, for one epoch.

6.3 Staged training

The reverse-training recipe of the second author builds a custom pretrained model in stages: a bidirectional Italian–English model trained on ParaCrawl, then the same model trained further on Sicilian forward-translations of that text, then a stage on back-translations of the NLLB Sicilian and finally the fine-tuning on *Arba Sicula*. Since NLLB-200 already provides a multilingual pretraining, we keep only the last two stages, which require data that he prepared for this

work (distinct from the back-translations of his earlier experiments mentioned in section 4). Starting from the monolingual Sicilian of NLLB (1,057,469 sentences), he back-translated it into English and Italian with his models, and scored each synthetic pair by the negative log-probability that its real side is a translation of the synthetic one (the lower, the better); since his models were trained on the literary standard, a standard Sicilian sentence usually obtains a lower score than a dialectal one. He did the same with the English side of the NLLB data (translated into Sicilian), with 941,391 Italian sentences of WikiMatrix (translated into Sicilian) and with the monolingual Sicilian of his sheets. Our second stage takes, for each of these files, the 100,000 pairs with the lowest score among those that pass a few filters: no unknown token, at least five words on each side, a source that is neither equal to nor a near-copy of the target (similarity below 0.9), no overlap with the test set, and a Sicilian side that does not end its words in -o more often than in -u (an inexpensive test against Italian passing as Sicilian, which tends to obtain the lowest scores). Together, the six lists cover the directions scn $\rightarrow$ en, en $\rightarrow$ scn, it $\rightarrow$ scn and scn $\rightarrow$ it, and 100,000 natural it-en pairs of WikiMatrix are added in both directions; the adapter is trained on them for one epoch. Finally, the third stage continues the same adapter on the curated set of section 3.4 in the four directions scn $\leftrightarrow$ en and scn $\leftrightarrow$ it, for two epochs.

Furthermore, the second author proposed not to discard the pairs with the worst scores, which contain the dialectal, Italian and Corsican text that the second stage filters out, but to spend them in a first, coarse stage, which spreads the output probabilities across the dialects and leaves broad peaks roughly in the right places, and to let the later stages sharpen them. This curriculum prepends to the two stages above one epoch on the 200,000 worst-scoring pairs of each file (with at least three words on each side and no filter on the endings), and is otherwise identical to them, so that the two runs can be compared.

7 Evaluation

We score the systems with sacreBLEU [21], i.e., with BLEU [17] (tokenization 13a) and chrF [20], on the frozen literary test set of section 3.4. We do not use the FLORES test set of NLLB for scn-en, since its Sicilian follows the orthography introduced after 2017 and comes from a different, non-literary domain, and is therefore not representative of the *Arba Sicula* material; FLORES-200 devtest is used only for it $\leftrightarrow$ en, which our test sets do not cover. Sockeye systems are scored in tokenized space, as the upstream system was, whereas NLLB systems are scored on raw text, with the Sicilian side normalized as in training for the staged runs; the two groups of rows of Table 2 are thus not strictly comparable (on raw text, the floor obtained by copying the source is 5.27 BLEU). Training was done on an 8-core CPU for the Sockeye models and on a single Colab GPU for the NLLB adapters. We report no running times, since only the quality of the translation is of concern here.

8 Results

Table 2 reports the results on the frozen test set of 1,000 literary pairs. In the table, “floor” is the score of the untranslated source, each row beginning with “+” adds one component to the row above it, “raw” means that NLLB receives the Sicilian text without normalization, and a dash marks a direction on which the model was not trained or evaluated. Note that the Sockeye baseline barely exceeds the floor (5.54 against 5.27 BLEU), and that the three levers raise it only to 10.85. NLLB-1.3B, on the other hand, is much better even zero-shot (29.02 on scn $\rightarrow$ en), and the bidirectional LoRA adapter trained on the web set raises it to 31.21, while en $\rightarrow$ scn improves from 9.89 to 19.03. Adding the 7,498 back-translated pairs gives 31.66 on scn $\rightarrow$ en; this small improvement is consistent with the size of the in-domain monolingual pool, which is

Table 2: Results on our test set (BLEU / chrF). The upper rows are trained on the web set, the staged rows on the back-translations and on the curated set (cf. section 6.3).

model	scn→en	en→scn
floor (copy source)	5.27 / 25.40	–
Sockeye baseline	5.54 / 28.28	–
+ lever B (tok. + desinences)	7.24 / 29.52	–
+ more data	9.79 / 33.82	–
+ lever D (lemma source factors)	10.85 / 35.22	–
NLLB-200 1.3B, zero-shot (raw)	29.02 / 55.23	9.89 / 41.12
LoRA, bidirectional	31.21 / 56.77	19.03 / 49.91
+ back-translation (7,498 pairs)	31.66 / 57.25	–
staged (back-translations, then curated)	32.40 / 58.19	21.72 / 52.79
curriculum (coarse stage first)	33.01 / 58.62	21.78 / 52.94

two orders of magnitude smaller than the back-translations used by the staged training.

Staged training improves both directions, with 32.40 BLEU on scn→en and 21.72 on en→scn; the gain over the bidirectional adapter is 1.19 BLEU on scn→en and 2.69 on en→scn, which may be due to the back-translations having been selected for the quality of their Sicilian, i.e., of the target side of en→scn. We remark that this comparison mixes two changes, since the staged run replaces the web set with the curated one as well as adding the back-translations; we have not trained the curated set alone. Prepending the coarse stage of section 6.3 raises scn→en to 33.01 BLEU (58.62 chrF), 0.61 more than the staged run, while en→scn is essentially unchanged (21.78 against 21.72). Repeated runs of the bidirectional adapter differ by up to about 0.2 BLEU on scn→en and 0.5 on en→scn, and therefore the first difference is probably real and the second is within the noise. Of course, the curriculum also trains for one more epoch on about 826,000 pairs, and hence we cannot tell whether its benefit comes from the dialectal text itself or from the longer training.

Italian directions. Italian is closely related to Sicilian and has much more data, and NLLB supports it↔scn natively. On the web set, we hold out a frozen it–scn test set of 1,000 pairs from the WikiMatrix data, evaluate NLLB-1.3B zero-shot, and then fine-tune the four directions (scn↔en and it↔scn) jointly into one multilingual adapter, which uses Italian as a bridge in the sense of Johnson et al. [10] and Fan et al. [5]. Table 3 reports the results as BLEU / chrF. As one would expect for a close, high-resource neighbor on in-domain WikiMatrix text, the Italian pairs are by far the “easiest”: scn→it reaches 43.70 BLEU, and it→scn improves from 17.61 to 27.48. However, the multilingual adapter loses somewhat on scn↔en (30.97 and 17.85 BLEU, against 31.21 and 19.03 for the bidirectional adapter of Table 2); hence, on the web set the Italian bridge brings no improvement to the English directions.

Forgetting of the untrained directions. Since the adapter is trained only on the directions that involve Sicilian, one may ask what happens to the others. On FLORES-200, the trilingual adapter lowers it→en from 34.40 to 27.18 BLEU and en→it from 29.43 to 28.76 (Table 3), i.e., it forgets a direction on which the base model was good and on which it was never trained. This is the transfer from high- to low-resource languages described by Johnson et al. [10], and it limits the use of the adapter as a general multilingual model; the translation between Sicilian and the other languages of NLLB (e.g., Spanish, French or Portuguese) is probably affected as well, although we have not measured it.

Table 3: Trilingual adapter on the web set, zero-shot and after fine-tuning (BLEU / chrF). The scn-en rows use our test set, the it-scn rows the WikiMatrix one, and the it-en rows the FLORES-200 devtest, on which the adapter was not trained.

direction	zero-shot	multilingual FT
scn→en	29.02 / 55.23	30.97 / 56.63
en→scn	9.89 / 41.12	17.85 / 48.82
it→scn	17.61 / 44.77	27.48 / 51.66
scn→it	42.20 / 59.04	43.70 / 59.89
it→en	34.40 / 62.61	27.18 / 57.94
en→it	29.43 / 58.63	28.76 / 58.06

Table 4: Published results of the second author against ours (BLEU). His results are on his test set, ours on the test sets of section 7; hence, the comparison is indirect (cf. section 8).

direction	Wdowiak			ours (NLLB-200 1.3B)			
	baseline	many-to-many	reverse	zero-shot	web	staged	curriculum
en→scn	25.1	35.0	45.1	9.89	19.03	21.72	21.78
scn→en	29.1	36.8	48.6	29.02	31.21	32.40	33.01
it→scn	–	36.5	61.4	17.61	27.48	–	–
scn→it	–	30.9	62.9	42.20	43.70	–	–
it→en	–	–	48.2	34.40	27.18	–	–
en→it	–	–	46.7	29.43	28.76	–	–

Relation to the state of the art. Our results remain below the best published ones, as Table 4 shows direction by direction. Its first three columns are the results of the second author: the baseline and the larger many-to-many model of Wdowiak [35] (the latter trained with back-translations and, for the Italian directions, with the trilingual textbook exercises), and the third stage of his reverse-training system [36], all measured on his test set, which was drawn by hand from the issues AS38 and AS39 of the journal. The other columns are ours: NLLB-1.3B zero-shot, the adapter trained on the web set (the bidirectional one for scn↔en, the trilingual one for the other directions), and the two staged runs of section 6.3, which were evaluated on scn↔en only. Our scn-en rows use our frozen test set, the it-scn rows the WikiMatrix one and the it-en rows the FLORES-200 devtest, and a dash marks a direction that was not reported. Since the two test sets come from similar material but differ, this is not a direct comparison; yet, the gap is too large to be due to the test set only, since we exceed his baseline on scn→en (33.01 against 29.1) and remain below it on en→scn (21.78 against 25.1). Also, the shape of the results differs. His model learned Sicilian by adapting an Italian-English one, and its scores fall into three nearly equal groups (from English, into English and between Romance languages); ours are uneven, with en→scn the weakest and scn→it the strongest, which is consistent with a model that learned Sicilian as one of 200 languages. Furthermore, the parts of his recipe that could be applied on top of NLLB give less than one might expect: orthographic normalization adds about 0.4 BLEU, the Italian bridge slightly lowers scn↔en, and the staged training on his back-translations and on the curated set adds 1.19 BLEU on scn→en and 2.69 on en→scn. The first stage of his recipe, i.e., the adaptation of an Italian-English model to Sicilian, is the part that we have not reproduced, and it may account for much of the remaining gap.

9 Conclusion and Future Work

We have rebuilt, on maintained software and without Perl, the path from the PDFs of *Arba Sicula* to a Sicilian–English NMT system, i.e., the extraction, alignment and cleaning of the parallel text, the tokenization and desinence-biased segmentation of the original system with an added lemma source factor, and the adaptation of NLLB-200 with LoRA, trained in stages on the back-translations of the second author and on the curated literary text. On our test set the best model, i.e., the curriculum proposed by the second author, reaches 33.01 BLEU on scn→en and 21.78 on en→scn, above the published baseline of the original system in the first direction but below it in the second, and well below the reverse-training results in both; its coarse stage on the badly scored back-translations helps the translation from Sicilian, but not the one into it.

Several extensions are open. In particular, the aligner could merge the short verses of poetry, as simulated in section 4.1, and could collect the Sicilian left unaligned as well as the English, which would enlarge the in-domain pool for back-translation. Also, the training data could be weighted by quality, e.g., favoring the issues published after 1998 or the texts written or translated by Cipolla, whose style varies less than that of the whole journal (the hard part being to identify them, since the issues carry no such metadata). If the back-translations of the web Sicilian turn out to be of little use, forward-translations produced by a small model trained on the literary text are an alternative for the pretraining stage. Furthermore, the forgetting of the untrained directions could be limited by mixing into the adaptation a large multi-parallel corpus (e.g., Bible translations in many directions), at the price of a much longer training. It would also be interesting to compare the encoder representations of translated sentences under bidirectional and unidirectional training, and at different depths, as suggested by the second author, and to measure how much the textbook exercises contribute. Finally, the same pipeline may be applied to varieties close to Sicilian, such as Calabrese and Salentino, whose written texts show the same orthographic diversity.

Our code for the pipeline and the evaluation is publicly available, together with a small serving application (a web interface and a Telegram bot, both built on a FastAPI service over the NLLB adapter); the text of the journal is under copyright, and the hand-aligned sheets are not public, so the datasets are not released.

Acknowledgments

We thank Gaetano Cipolla, *Arba Sicula* and Legas for granting permission to use the digital archive of *Arba Sicula*, and for 47 years of work on the journal.

References

- [1] Noëmi Aepli, Chantal Amrhein, Florian Schottmann, and Rico Sennrich. A benchmark for evaluating machine translation metrics on dialects without standard orthography. In Philipp Koehn, Barry Haddow, Tom Kocmi, and Christof Monz, editors, *Proceedings of the Eighth Conference on Machine Translation*, pages 1045–1065, Singapore, Dec 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.wmt-1.99. URL <https://aclanthology.org/2023.wmt-1.99/>.
- [2] Naveen Arivazhagan, Ankur Bapna, Orhan Firat, Dmitry Lepikhin, Melvin Johnson, Maxim Krikun, Mia Xu Chen, Yuan Cao, George F. Foster, Colin Cherry, Wolfgang Macherey, Zhifeng Chen, and Yonghui Wu. “Massively Multilingual Neural Machine Translation in the Wild: Findings and Challenges”. *arXiv*, 2019. URL <http://arxiv.org/abs/1907.05019>.

- [3] J. K. Bonner. *Introduction to Sicilian Grammar*. Legas, Brooklyn, NY, 2001.
- [4] Gaetano Cipolla. *Learn Sicilian, Mparamu lu sicilianu*. Legas, Mineola, NY, 2013.
- [5] Angela Fan, Shruti Bhosale, Holger Schwenk, Zhiyi Ma, Ahmed El-Kishky, Siddharth Goyal, Mandeep Baines, Onur Celebi, Guillaume Wenzek, Vishrav Chaudhary, Naman Goyal, Tom Birch, Vitaliy Liptchinsky, Sergey Edunov, Edouard Grave, Michael Auli, and Joulin Armand. “Beyond English-Centric Multilingual Machine translation”. *arXiv*, 2020. URL <https://arxiv.org/abs/2010.11125>.
- [6] Fangxiaoyu Feng, Yinfei Yang, Daniel Cer, Naveen Arivazhagan, and Wei Wang. Language-agnostic BERT sentence embedding. In *ACL*, 2022.
- [7] Kevin Heffernan, Onur Celebi, and Holger Schwenk. Bitext mining using distilled sentence representations for low-resource languages. *arXiv preprint arXiv:2205.12654*, 2022.
- [8] Felix Hieber, Tobias Domhan, Michael Denkowski, David Vilar, Artem Sokolov, Ann Clifton, and Matt Post. “Sockeye: A Toolkit for Neural Machine Translation”. *arXiv*, 2017. URL <http://arxiv.org/abs/1712.05690>.
- [9] Edward J. Hu, Yelong Shen, Phillip Wallis, et al. LoRA: Low-rank adaptation of large language models. *arXiv preprint arXiv:2106.09685*, 2021.
- [10] Melvin Johnson, Mike Schuster, Quoc V. Le, Maxim Krikun, Yonghui Wu, Zhifeng Chen, Nikhil Thorat, Fernanda B. Viégas, Martin Wattenberg, Greg Corrado, Macduff Hughes, and Jeffrey Dean. “Google’s Multilingual Neural Machine Translation System: Enabling Zero-Shot Translation”. *arXiv*, 2016. URL <http://arxiv.org/abs/1611.04558>.
- [11] Philipp Koehn and Rebecca Knowles. “Six Challenges for Neural Machine Translation”. *arXiv*, 2017. URL <http://arxiv.org/abs/1706.03872>.
- [12] Sneha Kudugunta, Ankur Bapna, Isaac Caswell, and Orhan Firat. Investigating multilingual NMT representations at scale. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 1565–1575, 2019. URL <https://aclanthology.org/D19-1167>.
- [13] Antonio Valerio Miceli Barone, Jindrich Helcl, Rico Sennrich, Barry Haddow, and Alexandra Birch. “Deep Architectures for Neural Machine Translation”. *arXiv*, 2017. URL <http://arxiv.org/abs/1707.07631>.
- [14] Vincenzo Mortillaro. *Nuovo Dizionario Siciliano-Italiano*. Lao, Palermo, 1876.
- [15] Vincenzo Nicotra. *Dizionario Siciliano-Italiano*. Bellini, Catania, 1883.
- [16] NLLB Team. No language left behind: Scaling human-centered machine translation. *arXiv preprint arXiv:2207.04672*, 2022.
- [17] Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. BLEU: a method for automatic evaluation of machine translation. In *ACL*, 2002. URL <https://aclanthology.org/P02-1040>.
- [18] Michele Pasqualino. *Vocabolario siciliano etimologico, italiano e latino*. Reale Stamperia, Palermo, 1785.
- [19] Giuseppe Pitrè. Saggio d’una grammatica del dialetto e delle parlate siciliane. In *Fiabe, novelle e racconti popolari siciliani*, volume 1 of *Biblioteca delle tradizioni popolari siciliane*. Luigi Pedone Lauriel, Palermo, 1875.

- [20] Maja Popović. chrF: character n-gram F-score for automatic MT evaluation. In *WMT*, 2015. URL <https://aclanthology.org/W15-3049>.
- [21] Matt Post. A call for clarity in reporting BLEU scores. In *WMT*, 2018. URL <https://arxiv.org/abs/1804.08771>.
- [22] Joseph F. Privitera. *Beginner’s Sicilian*. Hippocrene Books, New York, 1998.
- [23] Holger Schwenk, Vishrav Chaudhary, Shuo Sun, Hongyu Gong, and Francisco Guzmán. WikiMatrix: Mining 135m parallel sentences in 1620 language pairs from wikipedia. *EACL*, 2021.
- [24] Rico Sennrich and Barry Haddow. Linguistic input features improve neural machine translation. In *WMT*, 2016.
- [25] Rico Sennrich and Biao Zhang. “Revisiting Low-Resource Neural Machine Translation: A Case Study”. *arXiv*, 2019. URL <http://arxiv.org/abs/1905.11901>.
- [26] Rico Sennrich, Barry Haddow, and Alexandra Birch. “Improving Neural Machine Translation Models with Monolingual Data”. *arXiv*, 2015. URL <http://arxiv.org/abs/1511.06709>.
- [27] Rico Sennrich, Barry Haddow, and Alexandra Birch. “Neural Machine Translation of Rare Words with Subword Units”. *arXiv*, 2016. URL <http://arxiv.org/abs/1508.07909>.
- [28] Nitish Srivastava, Geoffrey Hinton, Alex Krizhevsky, Ilya Sutskever, and Ruslan Salakhutdinov. “Dropout: A Simple Way to Prevent Neural Networks from Overfitting”. *Journal of Machine Learning Research*, 15(56):1929–1958, 2014. URL <http://jmlr.org/papers/v15/srivastava14a.html>.
- [29] Brian Thompson and Philipp Koehn. Vecalign: Improved sentence alignment in linear time and space. In *EMNLP*, 2019.
- [30] Jorg Tiedemann. “Parallel Data, Tools and Interfaces in OPUS”. *Proceedings of the 8th International Conference on Language Resources and Evaluation*, 2012. URL <https://opus.nlpl.eu/>.
- [31] Antonino Traina. *Nuovo vocabolario siciliano-italiano*. G. Pedone Lauriel, Palermo, 1868.
- [32] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. “Attention Is All You Need”. *arXiv*, 2017. URL <http://arxiv.org/abs/1706.03762>.
- [33] Aastha A. K. Verma, Anwoy Chatterjee, Mehak Gupta, and Tanmoy Chakraborty. Multilingual language models encode script over linguistic structure. *arXiv preprint arXiv:2604.05090*, 2026. URL <https://arxiv.org/abs/2604.05090>.
- [34] Eryk Wdowiak. “Sicilian Translator: A Recipe for Low-Resource NMT”. In *arXiv preprint arXiv:2110.01938*, 2021. doi: 10.1007/978-3-031-10464-0_50. URL <https://arxiv.org/abs/2110.01938>.
- [35] Eryk Wdowiak. “A Recipe for Low-Resource NMT”. In *Intelligent Computing*, pages 739–746, Cham, Switzerland, 2022. Springer. doi: 10.1007/978-3-031-10464-0_50. URL https://doi.org/10.1007/978-3-031-10464-0_50.
- [36] Eryk Wdowiak. Reverse training strategy. https://www.doviak.net/pages/ml-sicilian/ml-scen_p07.shtml, n.d. Part of the series “Introduction to Sicilian NLP”. Accessed September 2026.

- [37] Biao Zhang, Philip Williams, Ivan Titov, and Rico Sennrich. “Improving Massively Multilingual Neural Machine Translation and Zero-Shot Translation”. *arXiv*, 2020. URL <https://arxiv.org/abs/2004.11867>.